

MCQ Masterclass — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

48.1 How to use this capstone

This lesson is a rapid-revision drill over the whole Computer Awareness course. It does not introduce new topics. It collects, unit by unit, the must-remember ladder (the facts worth one mark each) and the trap catalogue (the wrong beliefs the distractors rely on). Read each section as a pair: first the ladder sentence, then the trap that inverts it. Where a trap has a course registry ID (TRAP-01 and so on), the ID is given so the item can be traced back to the evidence.

Term	Meaning in this lesson
Must-remember ladder	The one-line fact per unit that directly answers a recall item.
Trap catalogue	The near-neighbour wrong belief the examiner plants as a distractor.
Near-neighbour contrast	The discipline of stating what a term is, what it is not, and its nearest confusion.

48.2 Memory and storage ladder

The speed order, fastest first, is registers, cache (L1, L2, L3), main memory (RAM), secondary storage (SSD/HDD), then tertiary or offline storage. RAM (Random Access Memory) is volatile primary memory: its contents are lost when power goes off. ROM (Read-Only Memory) is non-volatile and read-only in normal operation; it holds firmware such as the BIOS startup instructions. Cache is faster than main memory and holds recently and frequently used data; it keeps copies of data that live in RAM and is therefore not a backup. The CPU works directly with main memory and does not access secondary storage directly.

Definition (Volatile vs non-volatile). *Volatile memory loses its contents without power (RAM, cache, registers). Non-volatile memory retains its contents without power (ROM, SSD, HDD, magnetic tape).*

The traps: RAM is non-volatile (TRAP-01), ROM allows normal read and write (TRAP-02), and the CPU accesses secondary storage directly (TRAP-03) are all false. Cache being slower than RAM, or cache being a backup, inverts the same ladder.

48.3 Hardware, software and device ladder

A device driver is software, and the operating system is system software (TRAP-10). Open-source means the licence permits viewing, changing, and sharing the source; it is not the same as freeware (free of cost) or shareware (trial) (TRAP-23). The firmware that initialises the hardware and

loads the operating system is the BIOS, Basic Input/Output System (TRAP-24). The control unit controls the operation sequence and data flow. Input devices in the exam's drill set are the light pen, punch-card reader, digitizer tablet, and scanner; output devices are the plotter, drum printer, and sound card (TRAP-12). MICR reads magnetic-ink characters, OCR reads printed text, and OMR reads marks (TRAP-25). Large engineering drawings call for a plotter; ICs arrived in the third generation and microprocessors in the fourth (TRAP-26). The smallest unit of data is the bit; 8 bits make one byte (TRAP-27).

Example 48.3.1. *An item lists plotter, scanner, sound card, and digitizer tablet and asks which is input. Trace the drill: plotter is output, sound card is output, scanner is input, digitizer tablet is input. Any option calling the plotter an input device is the planted distractor.*

48.4 Operating system ladder

The operating system is the layer that provides a user-friendly interface between the user and the hardware. While it runs, it is resident in primary storage (main memory). Multitasking means running more than one program at a time. Android is the open-source mobile OS of the syllabus. These four sentences answer four distinct direct-recall items, so memorise them verbatim.

48.5 MS Word, Excel, Access and PowerPoint ladder

In MS Word, F7 runs spell check and Mail Merge combines letters with an address database (TRAP-29). A Section Break, not a Page Break, enables different headers and footers in different parts of a document (TRAP-13). In MS Excel, every formula must start with =; \$A\$1 is an absolute reference that does not shift when copied; COUNTIFS counts rows meeting several criteria, and SUMPRODUCT with multiplication does equivalent multi-criterion counting (TRAP-14, TRAP-30). In MS Access, the Select query retrieves data on a condition, while Update, Append, and Delete modify data (TRAP-31). In PowerPoint, F5 starts the slide show from the beginning; Shift plus F5 starts from the current slide (TRAP-32). The string .png is an extension; PNG is the format — never conflate the two (TRAP-34).

Example 48.5.1. *An item asks which break allows different headers in the two halves of a document. The Page Break option is the trap: a Page Break only pushes content to a new page. Only a Section Break restarts headers and footers, so it is the correct choice.*

48.6 Internet, web and email ladder

The Internet is the global network of networks; the WWW (World Wide Web) is the system of linked pages served over it. An ISP — Internet Service Provider — provides the connection. A URL (Uniform Resource Locator) is the address of a resource, and DNS (Domain Name System) resolves names to IP addresses. IPv4 is 32-bit; IPv6 is 128-bit with a fixed 40-byte header (TRAP-18). In an email address, @ separates the user name from the domain name (TRAP-28). BCC hides recipients from each other while the sender always knows and To/Cc stay visible (TRAP-16). POP3 — Post Office Protocol version 3 — downloads mail to the local machine; IMAP keeps mail synced on the server (TRAP-17). Secure transmission uses SSL/TLS. Decimal 10 equals binary 1010.

48.7 Networking, security and cloud ladder

BGP is an inter-domain routing protocol used between autonomous systems; OSPF and RIP are intra-domain (TRAP-20). ICMP is a control and diagnostic protocol that carries no application data and uses no ports (TRAP-19). HTTP/3 uses QUIC over UDP, not TCP (TRAP-21). TCP provides ordering with no timing guarantee; UDP is connectionless. A virus needs a host program, a worm self-propagates across networks, and a Trojan poses as useful software. Encryption provides confidentiality while a digital signature provides authenticity and integrity (TRAP-15). In the IaaS responsibility model, the provider manages the infrastructure (networking and storage hardware) and the customer manages the OS, runtime, applications, and data (TRAP-22).

48.8 IT Act and e-governance ladder

The Information Technology Act, 2000 is India's primary cyber law. Section 43 covers unauthorised access and data damage (penalty and compensation), and Section 66 covers computer-related offences. The e-governance service models are G2C (Government to Citizen), G2B (Government to Business), G2G (Government to Government), and G2E (Government to Employees). The anchor programmes are Digital India, NeGP/e-Kranti, and the NeGD (National e-Governance Division). Keep the Section 43/66 pair and the G2C/G2B pair straight: 43 is unauthorised access, 66 is computer offences; G2C serves the citizen, G2B serves business.

Definition (G2x models). *G2C delivers services to citizens, G2B to businesses, G2G between government departments, and G2E to government employees.*

48.9 Exam retrieval and common confusions

Ten distinctions prevent most errors across the whole paper.

1. RAM is volatile; ROM is non-volatile and read-only (TRAP-01, TRAP-02).
2. Cache is faster than RAM and is not a backup; the CPU never touches secondary storage directly (TRAP-03).
3. Plotter, drum printer, and sound card are output; scanner, light pen, and digitizer are input (TRAP-12).
4. A device driver is software; BIOS boots the machine (TRAP-10, TRAP-24).
5. Section Break changes headers; Page Break does not (TRAP-13).
6. Formulas start with =; \$A\$1 is absolute; Select query retrieves (TRAP-30, TRAP-31).
7. F5 runs the show from the beginning; F7 checks spelling (TRAP-32).
8. IPv6 is 128-bit; @ separates user name and domain name (TRAP-18, TRAP-28).
9. POP3 downloads locally; BCC hides recipients from each other (TRAP-16, TRAP-17).
10. Signature is not encryption; BGP is inter-domain; HTTP/3 is QUIC over UDP; IaaS provider manages hardware only (TRAP-15, TRAP-20, TRAP-21, TRAP-22).

Example 48.9.1. *An item states three claims: cache is faster than RAM; POP3 downloads mail to the local machine; IPv6 addresses are 256-bit. The first two trace to the ladder and are true; the third inverts TRAP-18 (IPv6 is 128-bit) and is false. Eliminate any option that asserts all three.*

Exercises

1. State the memory speed order and the volatility of RAM and ROM.
2. Classify plotter, scanner, sound card, and digitizer tablet as input or output, with one line of reasoning each.
3. Distinguish Section Break from Page Break and F5 from F7.
4. Explain the meaning of =, \$A\$1, and COUNTIFS in one line each.
5. Contrast POP3 with IMAP and BCC with Cc.
6. State the roles of BGP, ICMP, and QUIC in one line each.
7. Expand G2C, G2B, G2G, and G2E, and name the Act and sections behind Sections 43 and 66.

Solutions

1. Registers, cache (L1/L2/L3), RAM, secondary storage, tertiary/offline. RAM is volatile; ROM is non-volatile and read-only in normal operation.
2. Plotter: output (draws); scanner: input (reads in); sound card: output (plays sound); digitizer tablet: input (captures strokes).
3. Section Break restarts headers/footers; Page Break only starts a new page. F5 starts the slide show from the beginning; F7 runs spell check.
4. = marks a formula; \$A\$1 fixes both column and row; COUNTIFS counts rows meeting several criteria.
5. POP3 downloads mail locally; IMAP syncs with the server. BCC hides recipients from each other; Cc stays visible to all recipients.
6. BGP routes between autonomous systems; ICMP carries control and diagnostic messages, not data; QUIC over UDP carries HTTP/3.
7. G2C: to Citizen; G2B: to Business; G2G: to Government; G2E: to Employees. Information Technology Act, 2000: Section 43, unauthorised access and data damage; Section 66, computer-related offences.